12TH BIOLOGY · CHAPTER 14 · FULL CHAPTER NOTES

Human Urinary System

Topic-wise short and long questions with answers from the complete chapter

14.1 HUMAN URINARY SYSTEM

Q.1 [Short] What is excretion? Which metabolic wastes are removed by the urinary system?

In biological terms, excretion is the process by which an organism eliminates its metabolic wastes. The urinary (excretory) system is responsible for excretion. The metabolic wastes include nitrogenous wastes, extra water and extra salts, etc.

Q.2 [Short] Which three essential roles does the urinary system perform at the same time?

1. **Nitrogenous waste removal:** It filters out urea, a nitrogenous waste of protein breakdown in the liver.
2. **Osmoregulation:** It maintains the "water-to-salt" balance of the blood by adjusting how much water is reabsorbed or excreted.
3. **pH and ion regulation:** It balances the levels of sodium, potassium and hydrogen ions to keep the blood slightly basic (around pH 7.4).

Q.3 [Short] Given the definition of excretion, why is the large intestine not classified as a major excretory organ?

Excretion is the removal of **metabolic wastes**, i.e., wastes produced by the chemical reactions (metabolism) inside the body cells, such as urea. The large intestine mainly removes faeces, which consist of undigested food that was never absorbed into the body cells or used in metabolism. Removal of undigested food is not excretion. That is why the large intestine is not a major excretory organ.

This is the "Critical Thinking" question of the textbook; the textbook does not print its answer.

Q.4 [Long] Outline the organs of the human urinary system with their location and function.

The human urinary system consists of two kidneys, two ureters, one urinary bladder and one urethra.

- **Kidneys:** They are located high in the abdominal cavity, one on each side of the vertebral column, between the 12th thoracic and 3rd lumbar vertebrae. Due to the position of the liver, the right kidney is slightly lower and smaller than the left one. The upper parts of both kidneys are partially protected by the 11th and 12th ribs.
- **Ureters:** These are about 25 cm long tubes. A ureter begins at the funnel-shaped renal pelvis of the kidney. Both ureters open into the urinary bladder. They transport urine from the kidneys to the urinary bladder.
- **Urinary bladder:** It is a hollow and distensible muscular organ located within the pelvic cavity. It serves as the urine reservoir.

- **Urethra:** It is a tube that carries urine from the urinary bladder to the outside of the body.

Q.5 [Short] Why is the right kidney slightly lower and smaller than the left kidney?

The liver lies on the right side of the abdominal cavity, just above the right kidney. Due to the position of the liver, the right kidney is slightly lower and smaller than the left one.

14.1.1 Structure of Kidney

Q.6 [Long] Describe the external and internal structure of the human kidney.

External structure

A kidney is a reddish-brown, bean-shaped organ. Each kidney has an adrenal gland on its top. The kidney and its adrenal gland are surrounded by a fibrous **renal capsule**. Each kidney has a convex and a concave side. A depressed area on the concave side is the **renal hilum** (hilus). At this point, the renal artery enters the kidney, and the renal vein and ureter leave it.

Internal structure

- **Renal cortex:** It is the outer portion. It makes up about one-third of the kidney's mass.
- **Renal medulla:** It is the inner portion. It makes up two-thirds of the kidney. It consists of a number of cone-shaped **renal pyramids**.
- **Renal pelvis:** It is a funnel-shaped structure in the centre of the kidney. Urine is collected in the renal pelvis and leaves the kidney through the ureter.
- **Nephrons:** Each kidney contains about one million microscopic tubular structures called nephrons. These are the functional units of the kidney.

Q.7 [Short] What is the renal hilum? Which structures enter and leave the kidney at this point?

The renal hilum (hilus) is a depressed area on the concave side of the kidney. At this point, the renal artery enters the kidney, and the renal vein and the ureter leave the kidney.

Q.8 [Short] How does the renal cortex differ from the renal medulla?

Renal cortex	Renal medulla
Outer portion of the kidney.	Inner portion of the kidney.
Makes up about one-third of the kidney's mass.	Makes up two-thirds of the kidney.
Cortical nephrons lie primarily in it; peritubular capillaries surround the convoluted tubules here.	Consists of cone-shaped renal pyramids; contains long loops of Henle and vasa recta.

Q.9 [Long] Explain the detailed structure of a nephron. Also differentiate the two types of nephrons.

Each kidney contains about one million microscopic tubular structures called **nephrons**. These are the functional units of the kidney. A nephron consists of two main portions: a renal corpuscle and a renal tubule.

1. Renal corpuscle

It consists of two associated structures:

- **Glomerulus:** It is a capillary network derived from an afferent renal arteriole.
- **Bowman's capsule:** It is a cup-shaped, double-walled sac that surrounds the glomerulus.

2. Renal tubule

It is a long tubule that leads away from the Bowman's capsule. It consists of three connected portions:

- **Proximal convoluted tubule:** It is the beginning portion and is highly coiled. It dips into the medulla.
- **Loop of Henle:** It is a U-shaped hairpin turn with two limbs. The descending limb continues toward the medulla. The ascending limb returns toward the cortex and reaches parallel to the renal corpuscle.
- **Distal convoluted tubule:** The ascending limb opens into it. It is highly coiled again.

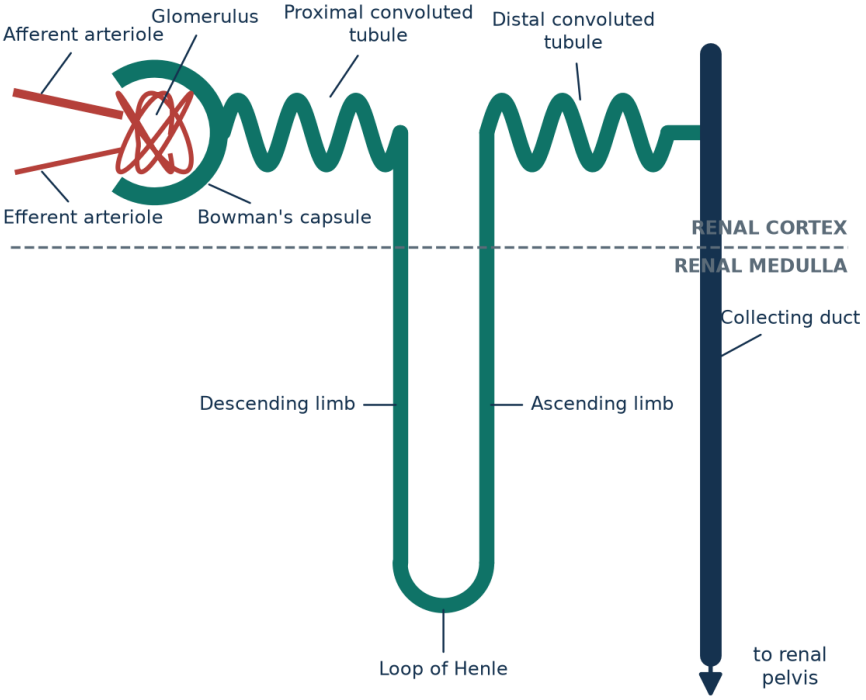
Collecting duct

The distal convoluted tubules of many nephrons drain into a single collecting duct. As these ducts descend through the medulla, they merge to form larger papillary ducts, which finally discharge into the renal pelvis.

Types of nephrons

Cortical nephrons	Juxtamedullary nephrons
They make the majority (about 85%) of nephrons.	They are fewer in number (the remaining, about 15%).
They are located primarily within the renal cortex.	They originate deep in the cortex, near the medulla.
They have relatively short loops of Henle.	They have long loops of Henle that extend deep into the renal medulla.

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Structure of a nephron (schematic)

Q.10 [Short] Why are nephrons called the functional units of the kidney?

All three steps of urine formation, i.e., glomerular filtration, selective reabsorption and tubular secretion, take place in the nephron. Each kidney has about one million nephrons, and the work of the kidney is the sum of the work of all its nephrons. So the nephron is the functional unit of the kidney.

Q.11 [Short] Trace the path of the filtrate from the Bowman's capsule to the renal pelvis.

Bowman's capsule → proximal convoluted tubule → descending limb of loop of Henle → ascending limb of loop of Henle → distal convoluted tubule → collecting duct → papillary duct → renal pelvis → ureter.

14.1.2 Blood Supply of the Kidney

Q.12 [Long] Describe the pathway of blood inside the kidney.

1. The **renal artery** enters the kidney at the hilum and divides into branches which give rise to **afferent arterioles**.
2. Each afferent arteriole leads into a capillary bed, i.e., the **glomerulus**.
3. Unlike most capillary beds of the body, which drain into venules, the glomerular capillaries drain into an **efferent arteriole**.
4. The efferent arteriole branches into **peritubular capillaries**. They form a dense network around the proximal and distal convoluted tubules in the renal cortex.
5. In juxtamedullary nephrons, the efferent arterioles also give rise to **vasa recta**. These are long, straight capillary networks in the medulla that run parallel to the loops of Henle.

6. After the exchange of materials, the capillaries join to form venules, which merge into larger veins and finally open into the **renal vein**.
7. The renal vein leaves the kidney at the hilum and drains into the inferior vena cava, which returns the filtered, “clean” blood to the heart.

The kidneys are highly vascularized organs. They receive approximately 20–25% of the total cardiac output.

Q.13 [Short] How is the glomerular capillary bed different from most other capillary beds of the body?

Most capillary beds of the body drain into venules. The glomerular capillaries, however, drain into an arteriole, i.e., the efferent arteriole. This efferent arteriole then branches again into a second capillary bed, the peritubular capillaries.

Q.14 [Short] What are vasa recta? What is their role?

Vasa recta are long and straight capillary networks in the medulla that run parallel to the loops of Henle. They arise from the efferent arterioles of juxtamedullary nephrons. The water that moves out of the collecting duct by osmosis enters the blood via the vasa recta.

14.1.3 Functioning of Kidney

Q.15 [Long] Explain glomerular filtration, selective reabsorption and tubular secretion as the events of kidney functioning.

Kidneys filter metabolic wastes and extra water from the blood and excrete them as urine. Urine formation has three major steps.

(i) Glomerular filtration

Glomerular filtration (ultrafiltration) is the first step in the formation of urine. It is a physical process. The efferent arteriole is narrower than the afferent arteriole. Due to this difference, the blood within the glomerular capillaries remains under relatively high pressure. This high pressure forces water, urea, glucose, amino acids, vitamins and various salts (Na^+, K^+, Cl^-) from the blood into the Bowman’s capsule.

About one-fifth (20%) of the fluid portion of the blood filters into the Bowman’s capsule and is called **glomerular filtrate**. Plasma proteins (e.g., albumin and globulins) and blood cells are too large to pass through the capillary walls, so they remain in the blood.

(ii) Selective reabsorption

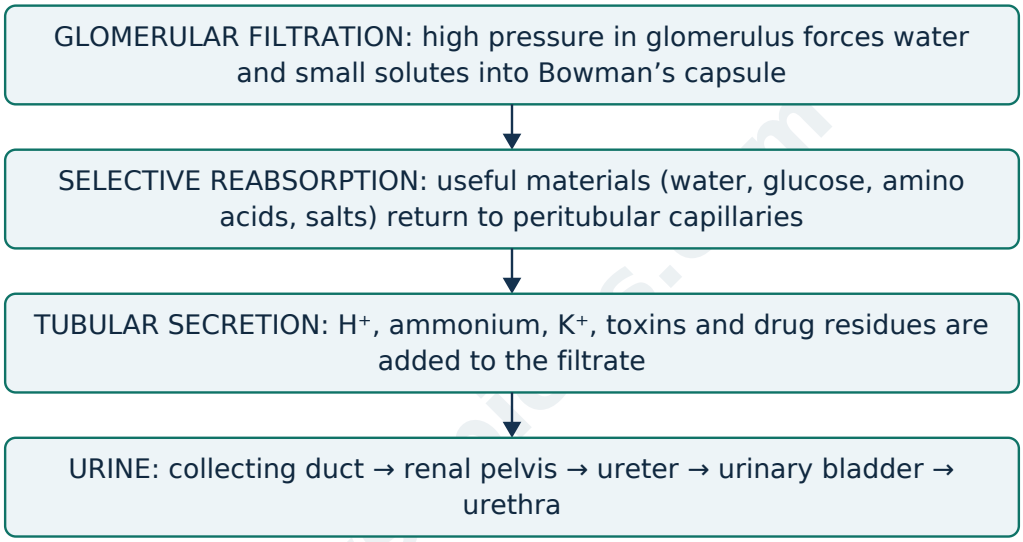
The glomerular filtrate contains not only wastes but also vital nutrients that the body cannot afford to lose. Selective reabsorption is the process by which useful materials are transported from the renal tubule back into the peritubular capillaries.

- **Proximal convoluted tubule:** Most reabsorption occurs here. About 75% of the water returns from the filtrate to the capillaries by osmosis. Glucose, amino acids and salts (sodium, potassium and calcium) are returned to the blood by active transport.
- **Descending limb of loop of Henle:** More water is reabsorbed here.

- **Ascending limb:** It is impermeable to water but actively transports salts out of the tubule.
- **Distal convoluted tubule:** Some water and salts are also reabsorbed here.
- **Collecting duct:** It is the final site for water reabsorption.

(iii) Tubular secretion

Tubular secretion is opposite to reabsorption. In this process, the epithelial cells of the distal convoluted tubule take some materials from the surrounding blood and secrete them into the filtrate. The secreted substances include hydrogen ions, ammonium, potassium ions and various metabolic toxins or drug residues (like penicillin). Through the tubular secretion of hydrogen ions, the kidneys help to maintain the acid–base balance (pH) of the blood.



Steps of urine formation

Q.16 [Short] **Why does the blood in the glomerulus remain under high pressure?**

The efferent arteriole, which carries blood away from the glomerulus, is narrower than the afferent arteriole, which brings blood into it. So blood enters the glomerulus more easily than it leaves. Due to this difference, the blood within the glomerular capillaries remains under relatively high pressure, and this pressure causes filtration.

Q.17 [Short] **Why are plasma proteins and blood cells not present in the glomerular filtrate?**

Plasma proteins (e.g., albumin and globulins) and blood cells are too large to pass through the walls of the glomerular capillaries. So they remain in the blood. Only water and small solutes such as urea, glucose, amino acids, vitamins and salts are filtered.

Q.18 [Short] **Why is selective reabsorption necessary?**

The glomerular filtrate contains not only wastes but also vital nutrients, such as glucose, amino acids, salts and a large amount of water. The body cannot afford to lose them. So these useful materials are transported from the renal tubule back into the peritubular capillaries.

Q.19 [Short] How are water and glucose reabsorbed in the proximal convoluted tubule?

In the proximal convoluted tubule, about 75% of the water returns from the filtrate to the capillaries by **osmosis**. Glucose, amino acids and salts (sodium, potassium and calcium) are returned to the blood by **active transport**, which uses energy.

Q.20 [Short] How does tubular secretion help to maintain the pH of the blood?

In tubular secretion, the cells of the distal convoluted tubule take hydrogen ions from the surrounding blood and secrete them into the filtrate. By removing the extra hydrogen ions in this way, the kidneys maintain the acid–base balance (pH) of the blood.

Q.21 [Short] How does selective reabsorption differ from tubular secretion?

Selective reabsorption	Tubular secretion
Materials move from the renal tubule back into the blood.	Materials move from the blood into the renal tubule (filtrate).
Useful materials are saved: water, glucose, amino acids, salts.	Unwanted materials are removed: hydrogen ions, ammonium, potassium ions, toxins, drug residues (penicillin).
Most of it occurs in the proximal convoluted tubule.	It occurs in the distal convoluted tubule.

Q.22 [Short] How is urine eliminated from the body? On which factors does the daily volume of urine depend?

From the collecting ducts, urine flows through the renal pelvis into a ureter, which leads to the urinary bladder. Muscular contractions of the bladder force the urine out of the body through the urethra. A normal adult eliminates 1 to 2 L of urine a day.

- The more water or hydrating fluids we consume, the higher is the urine output.
- Diets with high salt trigger water retention and decrease urine production.
- Physical exercise leads to water loss through respiration and perspiration, which generally decreases urine volume.
- In hot weather, the body loses more water through sweat, so the kidneys conserve water and produce a smaller volume of concentrated urine.

Q.23 [Short] How do kidneys work as osmoregulatory organs?

While doing the excretory function, kidneys also control the salt concentrations of body fluids. They form dilute urine when body fluids are hypotonic. They form concentrated urine when body fluids are hypertonic and there is dehydration.

14.2 REGULATION OF THE URINE CONCENTRATION

14.2.1 Counter-current Mechanism

Q.24 [Long] **How does the counter-current mechanism help in the formation of concentrated urine?**

For the reabsorption of water from the loop of Henle and the collecting duct, nephrons create and maintain a higher concentration of salts (NaCl) in the surrounding fluid. This is done by the counter-current mechanism. It depends on the different properties of the two limbs of the loop of Henle:

- The **descending limb** is permeable to water but not to salts.
- The **ascending limb** is impermeable to water but actively transports salts out of the tubule.

When the filtrate flows through the descending limb, water is reabsorbed into the surrounding fluid because the surrounding fluid has more salts. When the fluid flows through the ascending limb, sodium ions are actively transported from the filtrate to the surrounding fluid, and chloride ions follow the sodium ions. So the salt concentration of the surrounding fluid remains high. As urine passes down the collecting duct through this salty fluid, water moves out by osmosis and enters the blood via the vasa recta. In this way, concentrated urine is formed.

Q.25 [Short] **How do the two limbs of the loop of Henle differ in permeability?**

Descending limb	Ascending limb
It is permeable to water.	It is impermeable to water.
It is not permeable to salts.	It actively transports salts (sodium ions, followed by chloride ions) out of the tubule.
Water is reabsorbed from it into the salty surrounding fluid.	It keeps the salt concentration of the surrounding fluid high.

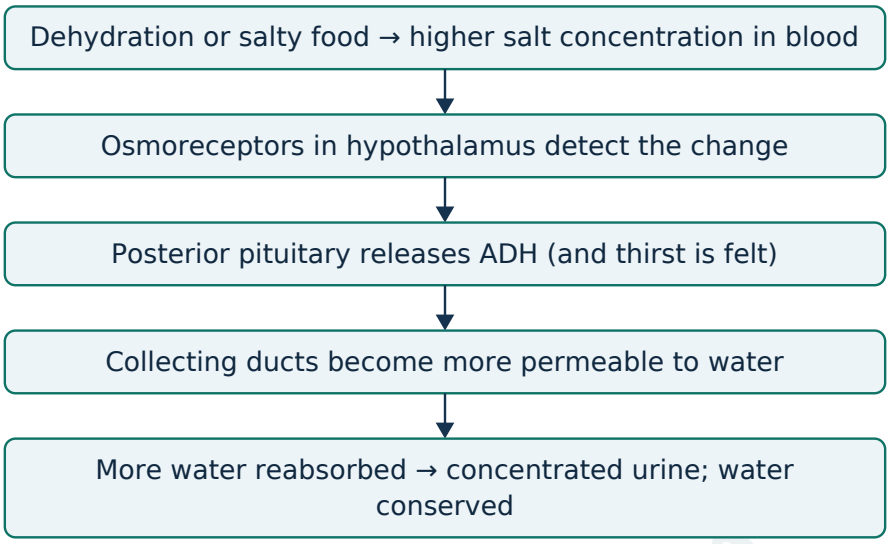
Hormonal Control of Kidney Functions

Q.26 [Long] **How does ADH control the amount of water in urine?**

ADH is the body's primary tool for water conservation. It is produced by the hypothalamus and secreted by the posterior pituitary gland.

- **When the body is dehydrated or salty food is taken:** The higher concentration of salts in the blood is detected by osmoreceptors in the hypothalamus. The hypothalamus signals the posterior pituitary to create a sensation of thirst and to release ADH. ADH acts on the walls of the collecting ducts and makes them more permeable to water. More water moves from the filtrate into the hypertonic fluid and then into the blood. So concentrated urine is produced and water is conserved.
- **When the concentration of salts in the blood falls:** The hypothalamus stops ADH secretion. The walls of the collecting ducts become less permeable to water.

Less water is reabsorbed and dilute urine is excreted. As a result, the blood volume and blood pressure decrease.



Action of ADH

Q.27 [Short] How does ADH relate to diabetes insipidus?

A person who cannot produce enough ADH due to pituitary damage constantly excretes a large volume of dilute urine, because the collecting ducts remain less permeable to water. This disorder is known as diabetes insipidus. Such a person is in danger of becoming severely dehydrated with very low blood pressure.

Q.28 [Short] How does ADH increase the blood pressure?

ADH increases the reabsorption of water, so the blood volume increases. This results in an increase in blood pressure. ADH also causes the blood vessels to constrict (narrow). This constriction further increases the blood pressure.

Q.29 [Short] How does aldosterone regulate the working of the kidney?

Aldosterone is the hormone responsible for regulating the sodium and potassium levels of urine. It is produced by the adrenal cortex (the outer layer of the adrenal gland present on the top of the kidney). It is secreted when the blood pressure drops or the sodium level in the blood is too low. It acts on the walls of the distal convoluted tubule and the collecting duct and increases the reabsorption of sodium ions from the filtrate into the blood. Due to the higher concentration of sodium ions in the blood, water also moves from the filtrate into the blood.

Q.30 [Short] How does aldosterone prevent hyperkalaemia?

Aldosterone also promotes the excretion of potassium ions in the urine. This helps to prevent hyperkalaemia, a condition in which the blood potassium level becomes too high and can lead to dangerous cardiac arrhythmias.

Q.31 [Short] How does ADH differ from aldosterone?

ADH	Aldosterone
Produced by the hypothalamus and secreted by the posterior pituitary gland.	Produced by the adrenal cortex.
Released when the salt concentration of the blood is high (dehydration, salty food).	Released when the blood pressure drops or the sodium level of the blood is too low.
Acts on the walls of the collecting ducts.	Acts on the walls of the distal convoluted tubule and the collecting duct.
Makes the walls more permeable to water, so more water is reabsorbed.	Increases the reabsorption of sodium ions; water follows the sodium ions into the blood.

Q.32 [Short] Compare the two capillary beds of the kidney.

Feature	Glomerular capillaries	Peritubular capillaries
Location	Within the Bowman's capsule	Around the renal tubules
Formed from	Afferent arteriole	Efferent arteriole
Pressure	High blood pressure	Low hydrostatic pressure
Function	Filtration: water and small solutes are forced out of the blood into the nephron	Reabsorption and secretion: nutrients (glucose, ions) and water are pulled back into the blood; specific toxins are secreted into the tubule

14.3 DISORDERS OF URINARY TRACT

14.3.1 Urinary Tract Infections

Q.33 [Short] What causes urinary tract infections (UTIs)? What are their symptoms and treatment?

Cause: UTIs are usually caused by bacteria, most commonly *Escherichia coli*. These bacteria enter the urinary system through the urethra and infect the bladder, ureters or kidneys.

Symptoms: A frequent and urgent need to urinate, pain or burning during urination, cloudy or strong-smelling urine, lower abdominal pain, and sometimes fever or chills.

Treatment: A course of antibiotics is given to kill the causative bacteria. Drinking plenty of water and urinating frequently also help to flush out the bacteria and reduce the symptoms.

14.3.2 Kidney Stones

Q.34 [Long] What are kidney stones? Explain their causes and treatment.

Urinary stones are hard, crystalline mineral materials that stick together to form small "pebbles" within the kidney or urinary tract. They may stay in the kidneys or travel out of the body through the urinary tract.

Causes

- **Calcium oxalate stones (70% of patients):** They are formed due to hypercalciuria (high calcium in urine) and hyperoxaluria (increased level of oxalate in urine).
- **Uric acid stones (10% of patients):** They are caused by hyperuricemia (increased amount of uric acid in blood).
- A high concentration of cysteine and phosphates in urine also causes kidney stones.
- Continuous dehydration of the body increases the chances of kidney stone formation.

Treatment

Treatment of kidney stones usually involves **lithotripsy**, i.e., the breaking of stones. There are two methods:

- **Extracorporeal Shockwave Lithotripsy (ECSWL):** An instrument called a lithotripter generates shockwaves from outside the patient's body. These shockwaves break the stone into small pieces. Most of the pieces then pass out through urine.
- **Percutaneous Nephrolithotripsy (PNL):** It is used for larger stones. A tube is inserted from the patient's back into the kidney. A small camera called a nephroscope is inserted through the tube to see the stone. Ultrasound equipment is then inserted to break the stone. The stone pieces are grasped with special equipment and pulled out of the kidney.

Q.35 [Short] How do doctors find the nature of a kidney stone from a urine test?

When the urine is acidic, the stone is of calcium oxalate. When the urine is alkaline, the stone is of calcium phosphate. When the urine is persistently acidic, the stone is of the uric acid type.

Q.36 [Short] How does ECSWL differ from PNL?

ECSWL	PNL
Extracorporeal Shockwave Lithotripsy.	Percutaneous Nephrolithotripsy.
Shockwaves are generated by a lithotripter from outside the body.	A tube is inserted from the patient's back into the kidney; a nephroscope and ultrasound equipment are passed through it.
The broken pieces pass out through urine.	The pieces are grasped with special equipment and pulled out.
Used for ordinary (smaller) stones.	Used for larger stones.

14.3.3 Kidney Failure

Q.37 [Short] What is kidney failure? How does chronic kidney failure differ from acute kidney failure?

Kidney (renal) failure is a condition in which the kidneys lose their ability to filter wastes from the blood and to regulate the fluid and salt balance of the body.

Chronic kidney failure	Acute kidney failure (kidney injury)
Develops over a longer period of time.	Occurs suddenly.
Irreversible.	Often resolves with treatment.
Caused by diabetes, high blood pressure or kidney disease (chronic infection, inflammation of glomeruli).	Caused by kidney infection, blockage in the urinary tract (kidney stones, bladder cancer, enlarged prostate), medications such as NSAIDs, and blood clots in renal arteries or arterioles.

Q.38 [Long] What is renal dialysis? Explain haemodialysis and peritoneal dialysis.

Renal dialysis is the treatment of renal failure in which the metabolic wastes are filtered from the blood by artificial methods. It has two general types.

1. Haemodialysis

In haemodialysis, the wastes are removed by circulating the blood outside the body through an equipment called a **dialyzer**. It is a type of filter and contains a semipermeable membrane made of cellulose. A catheter is inserted into an artery, usually in the arm. The blood is circulated through the dialyzer, which also contains a special fluid called **dialysate**. As the blood passes through the dialyzer, the dialysate attracts the wastes, water and salts out of the blood. They pass through the semipermeable membrane and dissolve in the dialysate. The clean blood then returns to the body through a second catheter inserted in a vein.

2. Peritoneal dialysis

Peritoneal dialysis uses the **peritoneal membrane** that lines the abdominal cavity. In this process, 1.5 to 2 litres of dialysate is infused into the patient's abdomen through a catheter. The dialysate is left in the abdomen for 4 to 6 hours. Waste products and excess fluid from the blood capillaries of the peritoneum pass into the dialysate. Then the dialysate is drained out of the abdomen and discarded. This process is repeated several times a day to remove enough wastes and excess fluid.

Q.39 [Short] What problems are associated with dialysis?

- Dialysis requires access to the bloodstream, which increases the risk of infections.
- Hypotension (low blood pressure) is a common complication of haemodialysis.
- Muscle cramps can occur during dialysis due to changes in salt levels.
- Dialysis can cause anaemia, which leads to fatigue and weakness.

- If too much fluid is removed too quickly, it can lead to dehydration or low blood pressure.
- Dialysis treatments can be emotionally challenging. Patients may feel anxious, depressed or overwhelmed by the demands of the treatment.

14.3.4 Kidney Transplant

Q.40 [Short] **What is a kidney transplant? To whom is it offered?**

Kidney transplant is a surgical procedure in which a person's non-functioning kidney is replaced with a healthy kidney from a donor. It is offered to individuals with end-stage kidney failure. The process requires careful evaluation, matching and monitoring to ensure the best results for both the recipient and the donor.

Q.41 [Short] **Who can be the potential donors for a kidney transplant?**

- **Related donor:** A person who is genetically related to the recipient, such as a parent, sibling or child, can be the best donor of a kidney.
- **Unrelated donor:** The donor is not genetically related to the recipient, such as a spouse, friend or co-worker. This type of donation is less common, but it can be successful if the blood group and tissue type of the donor and recipient match.
- **Deceased donor:** Many people register with hospitals for the donation of their organs after death. Their kidneys are carefully matched to recipients on the basis of blood group, tissue type and immune system compatibility.

Q.42 [Short] **Why are immunosuppressant medicines given after a kidney transplant? What problems are associated with transplant?**

The recipient's immune system may recognize the transplanted kidney as a foreign object and attack it. Immunosuppressant medications are given to prevent this rejection. The problems associated with kidney transplant are:

- **Organ rejection:** The recipient's immune system may recognize the transplanted kidney as a foreign object and attack it. This can occur even if the patient is taking immunosuppressant medications.
- **Infections:** Patients with a suppressed immune system are at a higher risk of developing infections after a transplant.
- **Side effects of medicines:** Immunosuppressant medications can cause high blood pressure, weight gain and an increased risk of infections and cancers.

Q.43 [Short] **Is kidney donation safe for the donor?**

Yes. Kidney donation can be a life-saving gift for someone with end-stage kidney disease, and it does not have significant long-term health risks for the donor. Studies have shown that kidney donors generally experience no negative impact on their overall health or lifespan.

Q.44 [Short] **Which are the important kidney transplant centres of Pakistan mentioned in the textbook?**

- **SIUT, Karachi** (Sindh Institute of Urology and Transplantation): Pakistan's first kidney transplant centre, established in 1985; one of the largest and most successful transplant centres in the world.
- **PKLI&RC, Lahore** (Pakistan Kidney and Liver Institute & Research Center): a tertiary-care hospital established in 2017 for medical care, transplantation and research.

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